

Confidence Transport for Adaptive Experimentation: Theory and a Controlled Audit

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Abstract

Model updates change the sensitivities used for allocation, while reward confidence may rely on sensitivities fixed when observations were collected. We give a finite-action confidence-transfer result under explicit pre-reward metric and solver certificates, yielding reward optimism and a regret bound on their joint validity event. Under fixed-reference and approximation-envelope assumptions, a corrected center supplies confidence without nonlinear optimizer convergence. In a nearly linear 29-parameter audit, all primary-policy float64 confidence and optimism diagnostics pass, yet the evaluated bound is vacuous and the multiplier policy has higher sample-mean pseudo-regret than its omission within each primary-horizon condition.

1 Adaptive model updates and confidence transport

Adaptive allocation may retrain a reward model, then select an ad or ranking action from predicted rewards and uncertainty; related neural-bandit methods use learned representations [2, 3]. Retraining changes replayed sensitivities and the Gram matrix. Collection-time confidence does not directly bound scores in the recomputed metric.

At round $t \leq T$, observe x_t , choose a_t from a fixed finite nonempty set $\mathcal{A}$, then observe r_t ; the confidence target is the conditional mean reward $\mu_t^*(a)$ used for allocation, not a treatment-effect estimand. Queries $q_t(a) \in \mathbb{R}^d$ have fixed coordinates and dimension. Let $\mathcal{H}_t^-$ contain the context and history before round- t randomization and $\mathcal{H}_t$ contain all pre-reward information. The metrics $\bar{V}_t, V_t$, radius $\beta_t \geq 0$, and θ_t when used are $\mathcal{H}_t^-$ -measurable (predictable); other score inputs and the selected action are finite and $\mathcal{H}_t$ -measurable. All validity events are measurable.

For $\lambda, \sigma > 0$, with I the d -dimensional identity, define reference geometry and widths by

$$\bar{V}_1 = \lambda I, \quad \bar{V}_{t+1} = \bar{V}_t + \sigma^{-2} q_t(a_t) q_t(a_t)^\top, \quad \bar{s}_t(a)^2 = q_t(a)^\top \bar{V}_t^{-1} q_t(a). \quad (1)$$

Define cumulative pseudo-regret and frozen information gain, with $\gamma_0 = 0$, as

$$R_T = \sum_{t=1}^T \left[\max_{a \in \mathcal{A}} \mu_t^*(a) - \mu_t^*(a_t) \right], \quad \gamma_T = \log \frac{\det \bar{V}_{T+1}}{\det \bar{V}_1}. \quad (2)$$

Let V_t be the exact current metric and C_t the algorithmic operator, both $d \times d$ symmetric positive-definite (SPD). Set $s_t(a)^2 = q_t(a)^\top V_t^{-1} q_t(a)$ and $s_{C,t}(a)^2 = q_t(a)^\top C_t^{-1} q_t(a)$.

Theorem 1 (Finite-action confidence transfer). Under this protocol, suppose $\|q_t(a)\|_2 \leq G < \infty$. Define E_{conf} as the event that, simultaneously for $t \leq T$ and $a \in \mathcal{A}$,

$$|\mu_t^*(a) - m_t(a)| \leq \beta_t \bar{s}_t(a) + b_t(a), \quad b_t(a) \geq 0. \quad (3)$$

Let $\bar{D}_t \geq 0$, $0 < \kappa_{-,t} \leq \kappa_{+,t} < \infty$, and $1 \leq \alpha_t < \infty$ be finite pre-reward factors. Define E_{cert} as simultaneous validity, for every $t \leq T$, of the Loewner comparisons

$$e^{-\bar{D}_t} \bar{V}_t \leq V_t \leq e^{\bar{D}_t} \bar{V}_t, \quad \kappa_{-,t} V_t \leq C_t \leq \kappa_{+,t} V_t, \quad (4)$$

and, for a nonnegative pre-reward width map,

$$s_{C,t}(a) \leq \hat{s}_t(a) \quad (a \in \mathcal{A}), \quad \hat{s}_t(a_t) \leq \alpha_t s_{C,t}(a_t). \quad (5)$$

Use this same realized width map in the score and selector:

$$U_t(a) = m_t(a) + \beta_t e^{\bar{D}_t/2} \sqrt{\kappa_{+,t}} \hat{s}_t(a) + b_t(a), \quad U_t(a_t) \geq \max_{a \in \mathcal{A}} U_t(a) - \xi_t, \quad \xi_t \geq 0. \quad (6)$$

On $E_{\text{conf}} \cap E_{\text{cert}}$, $U_t(a) \geq \mu_t^*(a)$ for every action and round, and pathwise

$$R_T \leq \sqrt{\left(\sigma^2 + \frac{G^2}{\lambda}\right) \gamma_T \sum_{t=1}^T \beta_t^2 \left[1 + \alpha_t e^{\bar{D}_t} \sqrt{\frac{\kappa_{+,t}}{\kappa_{-,t}}}\right]^2} + 2 \sum_{t=1}^T b_t(a_t) + \sum_{t=1}^T \xi_t. \quad (7)$$

If $\Pr(E_{\text{conf}}) \geq 1 - \delta$ and $\Pr(E_{\text{cert}}) \geq 1 - \delta_{\text{cert}}$, these conclusions hold with probability at least $1 - \delta - \delta_{\text{cert}}$.

Proof sketch. On the joint event, reference confidence, the inverse forms of (4), and all-action solver upper validity make the score optimistic. Approximate maximization, lower confidence at the played action, the reverse comparisons, and played-action sharpness give the roundwise coefficient in (7). Summation, Cauchy–Schwarz, and the frozen rank-one potential bound $\sum_t \bar{s}_t(a_t)^2 \leq (\sigma^2 + G^2/\lambda) \gamma_T$ yield the result. Thompson geometry supplies endpoint and path-length constructions for the first sandwich [4]; Section 2 supplies (3) without optimizer convergence.

2 A corrected center without optimizer convergence

Use a differentiable model mean $\mu_\theta(x, a)$ in fixed parameter coordinates $\theta \in \Theta \subseteq \mathbb{R}^d$, with Θ a convex smoothness region. For $z = (x, a)$, define $q_\theta(z) = \nabla_\theta \mu_\theta(z)$, $q_t(a) = q_{\theta_t}(x_t, a)$, $z_s = (x_s, a_s)$, and $q_s := q_{\theta_s}(x_s, a_s)$. The representation path $\theta_t \in \Theta$ is predictable. Fix before data collection $\theta^\circ \in \Theta$, $\|\theta^\circ\|_2 \leq S$, with $\mu_t^*(a) = \mu^*(x_t, a)$ and $\mu^*(z) = \mu_{\theta^\circ}(z) + m^\circ(z)$. Observed rewards obey $r_t = \mu_t^*(a_t) + \eta_t$ and $\mathbb{E}[e^{u\eta_t} \mid \mathcal{H}_t] \leq e^{u^2 \sigma^2/2}$ for every real u .

Supply nonnegative pre-reward bounds $\epsilon_{\text{lin}}(s)$ on the absolute Taylor remainder of $\mu_{\theta^\circ}(z_s)$ about θ_s , and $\epsilon_{\text{mis}}(s)$ for $|m^\circ(z_s)|$. Define the current actionwise envelopes $\epsilon_{\text{lin},t}(a)$, $\epsilon_{\text{mis},t}(a)$ analogously about θ_t ; the past sum below is known before selection. Set

$$y_s = r_s - \mu_{\theta_s}(z_s) + q_s^\top \theta_s, \quad \hat{\theta}_t^{\text{lin}} = \bar{V}_t^{-1} \sigma^{-2} \sum_{s < t} q_s y_s, \quad (8)$$

$$m_t^{\text{corr}}(a) = \mu_{\theta_t}(x_t, a) + q_t(a)^\top (\hat{\theta}_t^{\text{lin}} - \theta_t). \quad (9)$$

For $\delta \in (0, 1)$, self-normalization [1] and the Taylor envelopes give (3) with probability at least $1 - \delta$, simultaneously over rounds and actions, using $m_t = m_t^{\text{corr}}$,

$$\beta_t^{\text{corr}} = \sqrt{\gamma_{t-1} + 2 \log(1/\delta)} + \sqrt{\lambda} S + \frac{1}{\sigma} \sqrt{\sum_{s < t} [\epsilon_{\text{lin}}(s) + \epsilon_{\text{mis}}(s)]^2}, \quad b_t(a) = \epsilon_{\text{lin},t}(a) + \epsilon_{\text{mis},t}(a). \quad (10)$$

The exact current squared-loss generalized Gauss-Newton (GGN) metric is $V_t = \lambda I + \sigma^{-2} \sum_{s < t} q_{\theta_t}(z_s) q_{\theta_t}(z_s)^\top$.

3 Controlled audit

We ran four policies in a nearly linear scaled-tanh bandit with five actions and 29 parameters. For normalized features $\|\phi(z)\|_2 \leq 1$, the mean is $\mu_{\theta, W}(z) = \sqrt{W} \tanh(\phi(z)^\top \theta / \sqrt{W})$. Exact realizability holds with $\|\theta^\circ\|_2 \leq R = 1$; parameters are projected to radius R , $\lambda = 1$, and Gaussian noise has $\sigma = 0.25$. For each horizon $T \in \{250, 500, 1000\}$ and target $D_{\text{target}} \in \{0.25, 0.5, 1, 2\}$, the predeclared rule sets

$$W = \left[\frac{4c_h R \sqrt{T} - 1}{\sigma \sqrt{\lambda} D_{\text{target}}} \right]^2, \quad c_h = \frac{4}{3\sqrt{3}}, \quad L_g = \frac{c_h}{\sqrt{W}}, \quad Q_t = \sum_{s < t} \|\theta_t - \theta_s\|_2^2, \quad D_{Q,t} = \frac{2L_g \sqrt{Q_t}}{\sigma \sqrt{\lambda}}. \quad (11)$$

For $u = \phi(z)^\top \theta$, $|u| \leq 1$, and $f_W(u) = \sqrt{W} \tanh(u/\sqrt{W})$, $0 \leq 1 - f_W(u)/u \leq u^2/(3W) \leq 1/(3W)$, using the continuous extension at zero; at $T = 1000$, the bound $1/(3W)$ ranges from 1.375×10^{-7} to 8.798×10^{-6} across the four conditions. Here L_g bounds the operator norm of the mean Hessian along replay segments. D_{target} equals $D_{Q,T}$ when every replay displacement is maximal ($2R$). Four policies, three separately run horizons, four targets, and 50 evaluation seeds give 2,400

trajectories. Ten disjoint tuning seeds selected shared optimizer settings by mean squared prediction error. Policies share contexts and per-action noise draws; their histories diverge. All use dense float64 Cholesky solves and exact argmax over the five actions, with failure level $\delta = 0.05$. The exact-arithmetic primary instantiation has $C_t = V_t$, $\kappa_{-,t} = \kappa_{+,t} = \alpha_t = 1$, and $\xi_t = 0$.

The transport-endpoint policy is a dense endpoint-certificate comparator. It uses $d_{\text{Th},t} = \max_i |\log \lambda_i(V_t, \bar{V}_t)|$, with λ_i the generalized eigenvalues of $(V_t, \bar{V}_t)$: the smallest common exponent for the first symmetric sandwich in (4) [4]. Both matrices are known before the reward, so the certificate is valid in exact arithmetic; its float64 value is not a verified enclosure. Each policy scores $m_t^{\text{corr}}(a) + \beta_t^{\text{corr}} w_t(a) + b_t(a)$ on its own history: Widths are $w_t = e^{D_{Q,t}/2} s_t$ (transport Hessian), $e^{d_{\text{Th},t}/2} s_t$ (transport endpoint), $\bar{s}_t$ (frozen reference), and s_t (naive current). The primary policy sets $\bar{D}_t = D_{Q,t}$; “Hessian” names the envelope argument. Let $\mathcal{D}_{\text{selected},t}$ be the exact Thompson-Finsler length of the SPD path that interpolates each replay parameter from θ_s to θ_t ; then $d_{\text{Th},t} \leq \mathcal{D}_{\text{selected},t} \leq D_{Q,t}$, and $\mathcal{D}_{\text{quad},t}$ is its 32-point Gauss-Legendre estimate at the declared checkpoints.

D_{target}	0.25	0.5	1	2
Transport Hessian mean R_{1000}	81.05	82.44	87.04	93.99
Transport endpoint mean R_{1000}	79.05	79.49	80.06	81.89
Frozen reference mean R_{1000}	79.09	80.13	80.25	81.67
Naive current mean R_{1000}	79.00	79.49	80.10	81.88
Median $D_Q/\mathcal{D}_{\text{quad}}$	1.88×10^6	9.77×10^5	4.82×10^5	2.51×10^5
Median trajectory-maximum d_{Th}	1.31×10^{-7}	5.23×10^{-7}	2.10×10^{-6}	8.74×10^{-6}
Median $e^{D_Q/2}$	1.02	1.04	1.09	1.18

Table 1: Primary horizon $T = 1000$. Regret means average 50 trajectories. The last three rows concern the primary policy: path ratios pool eligible seed-checkpoint pairs with $\mathcal{D}_{\text{quad}} > 10^{-12}$; endpoint distances are maximized within each trajectory before taking the median; inflation medians pool seeds and all rounds, not terminal rounds or run maxima.

Within each $T = 1000$ condition, the analytic multiplier raised sample mean pseudo-regret relative to the otherwise matched naive-current policy (Table 1); histories diverge after the first differing action, and the contrast is descriptive: D_{target} also changes W , so it is not a dose response. In each of the 12 horizon-target cells, 50/50 primary trajectories passed both all-action, all-round reference-confidence and optimism diagnostics, with cellwise exact 95% Clopper-Pearson interval [92.9%, 100%]. Base seeds are reused across cells. These tolerance-based float64 checks and quadrature values are diagnostics, not verified enclosures. Across the 12 primary cells, medians of per-trajectory ratios of the numerical right-hand side (RHS) of (7) to pseudo-regret range from 126.7 to 155.7. Every primary trajectory’s terminal RHS exceeds the $2T$ pseudo-regret cap from $|\mu^*(z)| \leq 1$.

4 Implications for adaptive experimentation

On the joint confidence-and-certificate event, the theorem gives optimism and a regret bound, but valid certificates need not improve allocation: in this nearly linear simulation, the analytic multiplier accompanies higher sample-mean pseudo-regret than its omission within each primary-horizon condition. The audit does not establish a uniform policy ordering or treatment-effect inference after adaptive assignment.

References

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